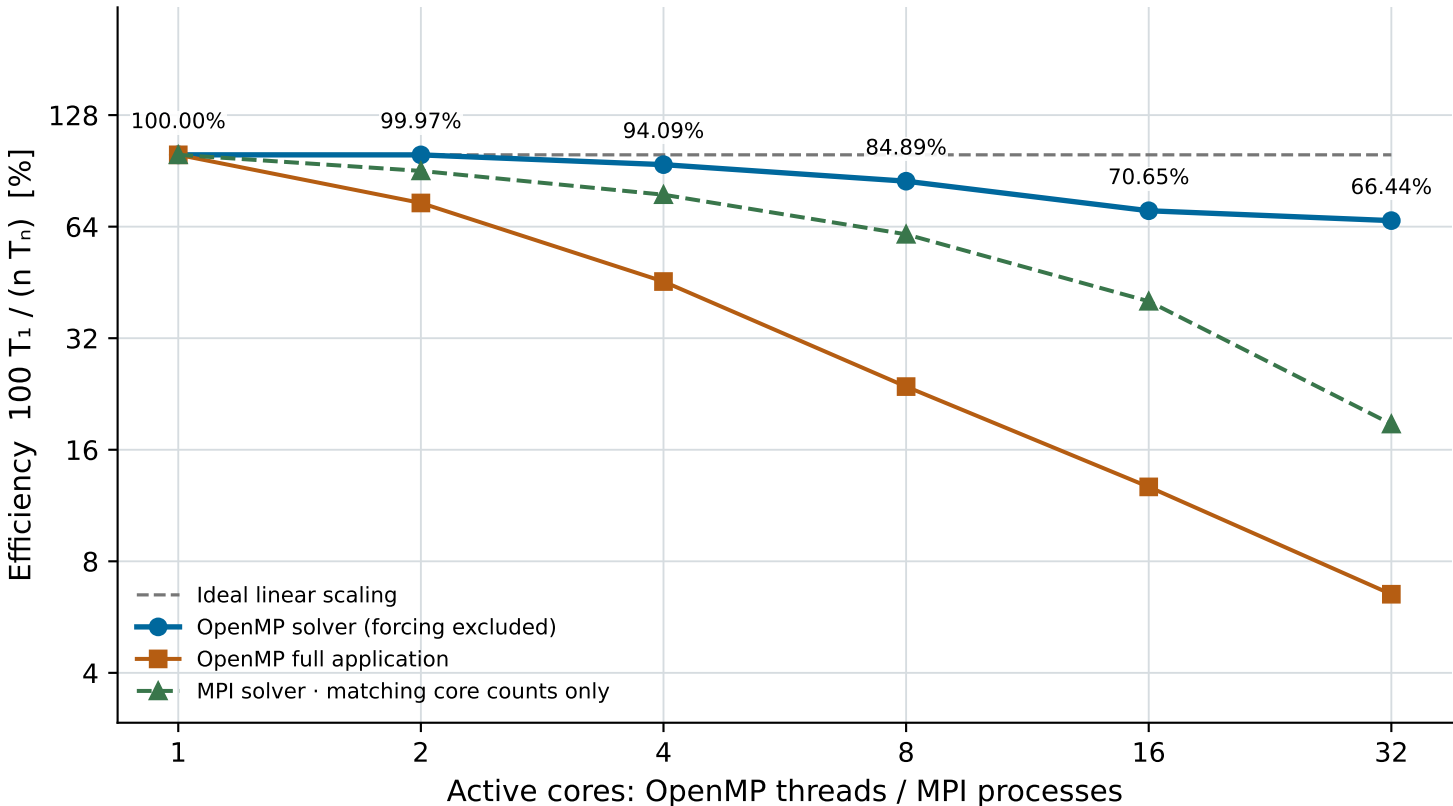


Sarno lite · one MPI process, varying OpenMP threads



Exclusive node · same executable, native forcing and six-hour workload

Separate solver baselines: OpenMP 10.927 s; MPI 10.562 s

One observation per point; no uncertainty bars. MPI and OpenMP use the same active core counts.